

Nikhil Krishna R D

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AI/ML Engineer | Data Analyst

AI/ML Engineer with production experience shipping ML pipelines, explainable models, and validated forecasting systems - not just notebooks. Built and rigorously validated an XGBoost + SHAP pipeline with walk-forward cross-validation, catching an overfit model before it shipped and correcting a mis-specified causality test. Deployed a ResNet-50 transfer learning pipeline (94% accuracy) via a live Flask inference app, and automation systems that eliminated 97% of manual reporting at Mahindra CAI. Oracle OCI Certified Data Science Professional · Published researcher (UGC CARE, 2025).

SKILLS

- **Machine Learning:** Python, XGBoost, SHAP, Prophet, ResNet-50, scikit-learn, Pandas, NumPy
- **MLOps & Deployment:** MLflow, Docker, FastAPI, Flask, CI/CD
- **Data Engineering:** SQL, PostgreSQL, MongoDB, pandera, Power BI
- **Tools:** Git, Playwright, Excel Automation, Gemini API (LLM)

WORK EXPERIENCE

RLabZ | Web Developer

Nov 2025 - Present

- Building the official website for an international academic conference at Rajagiri College (300+ expected attendees), owning frontend architecture through production deployment and CI/CD pipeline for iterative updates
- Collaborating with design and content teams to meet institutional branding standards

Kumaraguru College of Liberal Arts and Science | Automation and Platform Developer

Dec 2024 - Mar 2025

- Reduced event report generation from 1–2 hours to under 30 seconds via an algorithm-driven DOCX automation system for workshops, masterclasses, and field visits.
- Built a marklist processing platform for 1,400+ students across 12 departments, cutting processing time from 2+ hours to under 2 minutes with structured, analytics-ready outputs.
- Developed real-time result analysis dashboards with role-based access control for Admin, HOD, and Faculty users.

CAI Mahindra | Data Analyst Intern - Automotive Operations Analytics

Jul 2024 - Aug 2024

- Mapped ETBR (Enquiry→Test Drive→Booking→Retail) sales funnel for dealership network, identifying conversion drop-off patterns for operations leadership.
- Automated 15+ recurring Excel reports for Sales and Telecalling teams, cutting report generation time by 97% and eliminating manual analyst effort.
- Deployed a Flask-based internal KPI dashboard for self-serve sales metrics access, reducing ad-hoc requests to the analytics team

PROJECTS

Supply vs. Demand Drivers of India's CPI Inflation | github.com/rdnk2004/CPI-MPC

Jun 2026 - Jul 2026

- Analyzed 10 years of RBI Monetary Policy Committee decisions (60 meetings) against CPI data using STL decomposition and stationarity-corrected Granger causality, finding 5 of 8 rate hikes (62.5%) coincided with food-driven supply shocks
- Built an XGBoost + SHAP pipeline with walk-forward cross-validation; caught an in-sample overfit and correctly reported true out-of-sample skill ($F1 = 0.000$) rather than the misleading headline accuracy
- Independently validated model rationale against RBI's published minutes via LLM-based text comparison (57 meetings) and backtested a Prophet forecast's confidence intervals, exposing 60% actual vs. 90% claimed coverage

Nexus Task-Tracker | github.com/rdnk2004/nexus-tasktrack

Jan 2026 - Feb 2026

- Built a multi-user project management app with JWT authentication, Kanban-style task workflows, and enforced task lifecycle state rules to maintain data integrity
- Developed REST APIs using FastAPI and PostgreSQL with clean separation of backend logic, input validation, and role-based task access
- Containerised and deployed using Docker and Nginx for production-ready access with zero manual setup for new environments

Anomaly Detection Using ResNet-50 | github.com/rdnk2004/anomaly-detection

Dec 2024 - Feb 2025

- Designed and trained a ResNet-50 transfer learning pipeline for industrial defect classification, achieving **94% accuracy on hazelnut defects and 87% on bottle defects** across a multi-class imbalanced dataset.
- Applied production ML techniques - data augmentation, early stopping, and learning rate scheduling - to prevent overfitting; deployed inference via a Flask web application with real-time defect visualization.

EDUCATION

Master of Science, Computer Science (Data Analytics)

Jun 2025 - Mar 2027

Rajagiri College of Social Sciences (GPA: 8.02/10)

Bachelor of Science, Data Science

Aug 2022 - Jun 2025

Kumaraguru College of Liberal Arts and Science (GPA: 9.022/10)

- **Achievements:** University 3rd Rank, Best Student Award (Data Science), Mahatma Gandhi Merit Scholarship (3 Times)

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 – Certified Data Science Professional: Oracle University
- Applied Data Science with Python – Level 2: IBM
- Database Management Essentials (Honors): University of Colorado (Coursera)

PUBLICATIONS

- Nikhil Krishna R D.AIEDUDOCs: AN INTELLIGENT FRAMEWORK FOR AUTOMATED ACADEMIC DOCUMENTATION AND ASSESSMENT ANALYTICS. *Madhya Bharti* (UGC CARE), May 2025. Deployed across 150 faculty/12 departments; reduced documentation time by 87%, improved format consistency by 92%, and achieved 4.5/5 user satisfaction ($n=72$).